The Convolution Process

Original image



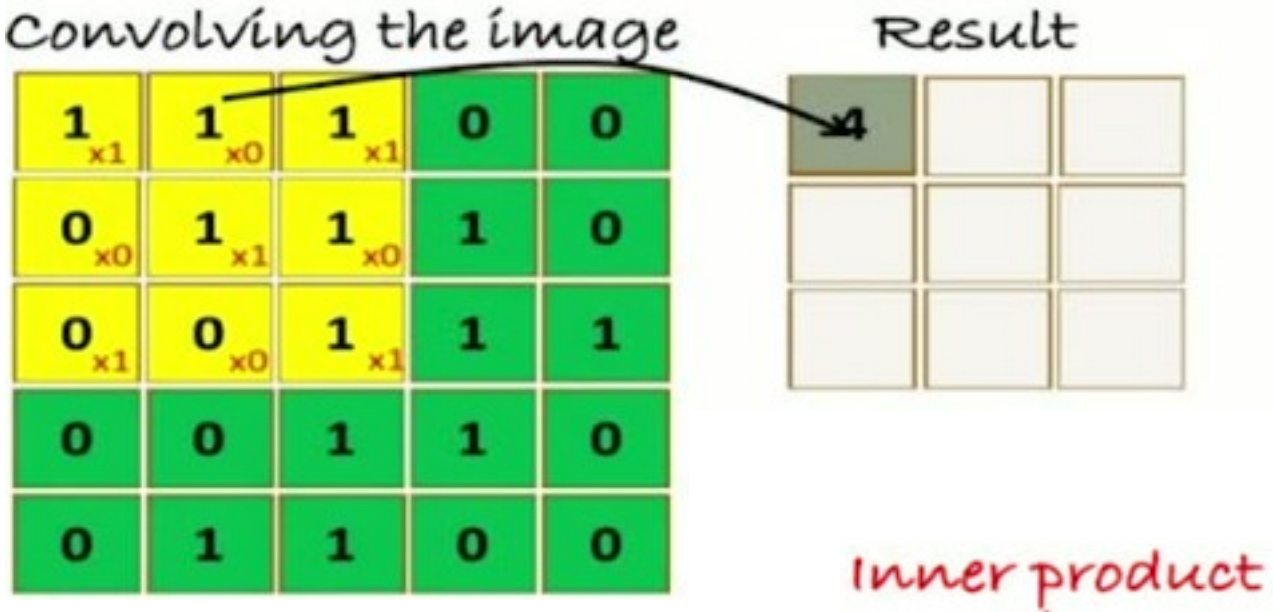
Convolutional filter 1

1	0	1
0	1	0
1	0	1

a=1, b=1

Image: I (x,y)

Filter: h (-a:a , -b:b) *(odd dimension)*



$$I(x,y) * h = \sum_{i=-a}^{a} \sum_{j=-b}^{b} I(x-i,\ y-j) \cdot h(i,j)$$

Scan the original image using the 3x3 filter

Move the filter ONE pixel (Right, Left, Up, Down) and Re-Calculate